

Functional Forecasting of Turkish Electricity Load Curves

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Turkish electricity market: actors and daily operation

Bilateral contracts → DAM → IDM → BPM → Settlement

Key actors

- **TEİAŞ** — TSO; maintains real-time system balance
- **EPIAŞ** — market operator; runs DAM and IDM; performs settlement
- **Generators** — submit supply offers
- **Suppliers/traders** — buy electricity, serve consumers
- **Distribution cos.** — regulated local networks
- **Consumers** — households, firms, eligible consumers

DAM ($D - 1$)	Hourly supply and demand cleared by EPIAŞ ⇒ MCP
IDM (<i>after DAM</i>)	Participants revise positions as wind/solar or demand shifts
BPM (<i>real time</i>)	TEİAŞ activates balancing bids to keep system in balance ⇒ SMP
Settlement	EPIAŞ compares scheduled and metered quantities; deviations settled financially

Financial motivation to improve the official load forecast

$e_t = L_t - \hat{L}_t^{\text{off}}$ ($e_t > 0$: underprediction, system short; $e_t < 0$: overprediction, system long).

Stylised imbalance cost:

$$\Delta C_t \approx (\text{SMP}_t - \text{MCP}_t) e_t$$

SMP is not always above MCP. The issue is that SMP is **volatile**: set in real time under scarcity or surplus, it can diverge substantially from MCP.

Larger forecast errors $\Rightarrow$ greater exposure to an unfavourable (SMP – MCP) spread.

better forecast $\Rightarrow$ smaller imbalance

$\Rightarrow$ less BPM exposure $\Rightarrow$ lower expected cost

Benchmark: TEİAŞ's official day-ahead load forecast (*Yük Tahmin Planı*), available on the EPIAŞ Transparency Platform.

Forecasting demand accurately reduces the amount of electricity that must be corrected unexpectedly in real time.

The problem: official forecast persistently underpredicts

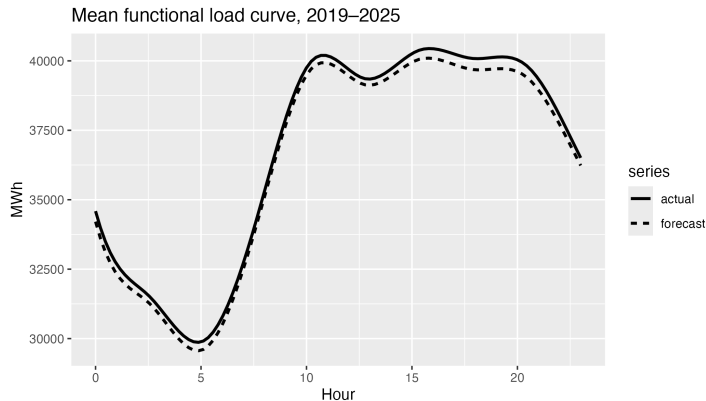
Why does the official underpredict?

Year	Official bias (MWh/hr)	Our bias (MWh/hr)
2023	+344	+176
2024	+585	+34
2025	+472	+98
Avg	+467	+103

Positive = actual > forecast (underprediction).

- **Asymmetric loss:** over-procurement is politically and contractually more costly than under-procurement — optimal forecast under asymmetric loss sits below the conditional mean.
- **Model staleness:** official models update infrequently; structural demand growth driven by industrialisation outpaces recalibration.
- **Strategic feedback:** the official forecast acts as a coordination device for generators; this can turn it into a focal point rather than an accurate estimate.

The object: a curve, not 24 numbers



$$Y_d(h), \quad h = 0, \dots, 23$$

- One coherent daily trajectory
- Ramp, plateau, and evening peak are *features of the whole curve*, not isolated hours
- Official (dashed) sits below actual at every hour
- **Goal:** forecast tomorrow's full profile $Y_{d+1}(\cdot)$

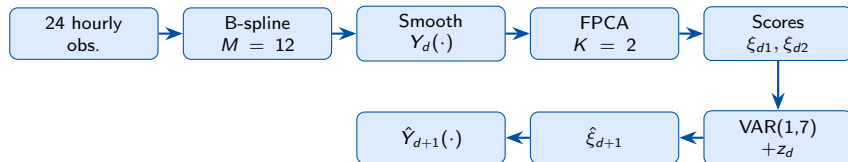
Research question

Main RQ: *Can a functional model of the daily load curve statistically outperform the official TEİAŞ day-ahead forecast?*

- ① **Sub-Q1 (Representation).** Do B-spline smoothing and FPCA represent Turkey's daily load curves with negligible information loss? How many components suffice?
- ② **Sub-Q2 (Linearity).** Is the linear score equation adequate, or does a nonlinear model *outside* the FPCA setting suggest unexploited structure in the data?

Literature positioning is provisional, pending the systematic review. No prior FDA study of Turkish electricity demand has been located. FDA framework: (Ramsay and Silverman, 2005; Kokoszka and Reimherr, 2017; Hyndman and Ullah, 2007).

Method: the pipeline



$$Y_d(h) = \mu(h) + \sum_k \xi_{dk} \varphi_k(h) + \varepsilon_d(h)$$

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$$

$$\hat{Y}_d(h) = \hat{\mu}(h) + \hat{\varphi}_1(h) \hat{\xi}_{d1} + \hat{\varphi}_2(h) \hat{\xi}_{d2}$$

OLS, equation-by-equation. All z_d predetermined.
 FPCA: (Ramsay and Silverman, 2005; Kokoszka and Reimherr, 2017);
 functional forecasting: (Hyndman and Ullah, 2007)

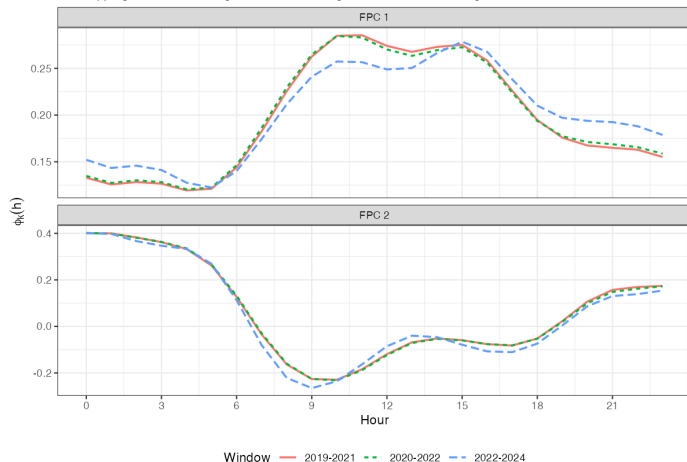
Block	Key content
Calendar	Weekend, annual sine/cosine
Holiday	Public holidays, bridge days
Load shape	Mean, peak, range, rolling avg.
Deriv.	Morning/evening ramp (DY_{d-1})

Rolling-origin / expanding window: all z_d strictly predetermined $\Rightarrow$ **no leakage**. Underpins all results.

FPCA: two scores capture the curve (Sub-Q1)

FPCA eigenfunctions across sub-periods

Overlapping lines = stable eigenfunctions; divergence = structural change



FPC	Var.	Cumul.
1	93.4%	93.4%
2	3.9%	97.3%
3	1.1%	98.4%
4	1.0%	99.4%

FPC 1 (level): nearly flat $\Rightarrow$ uniform lift/drop.

FPC 2 (shape): positive overnight/evening, negative daytime $\Rightarrow$ peak-vs-plateau contrast.

$K = 4$ adds $< 1\%$ variance; results unchanged.

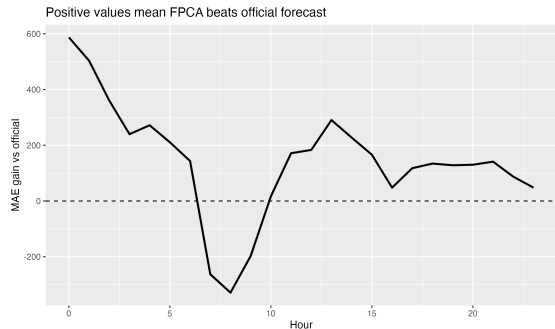
Static FPCA is a compromise between simplicity and dimension reduction, not a claim of optimality.

Main results: building the model (Main RQ)

Model	MAE	MAPE	vs. off.	DM stat	
Seasonal naïve Y_{d-7}	1912	5.13%	−57.6%	+35.2	
Official (TEİAŞ)	1214	3.15%	ref.	—	
FPCA VAR(1,7) only	1443	3.76%	−18.9%	+22.9	
+ calendar / ramp	1238	3.26%	−2.1%	+2.6	
+ holiday / load-shape	1001	2.63%	+17.5%	−24.9	***
+ derivative features	980	2.56%	+19.2%	−27.4	***
+ residual correction	653	1.70%	+46.2%	−71.2	***

Negative DM stat = better than official; *** $p < 0.001$. **DM significant at 18 of 24 hours** (hour-by-hour; pooled p is secondary). MAE in MWh/hour. Evaluation: 1,096 days / 26,304 hourly errors, 2023–2025. DM test: Diebold and Mariano (1995); HLN small-sample correction: Harvey et al. (1997).

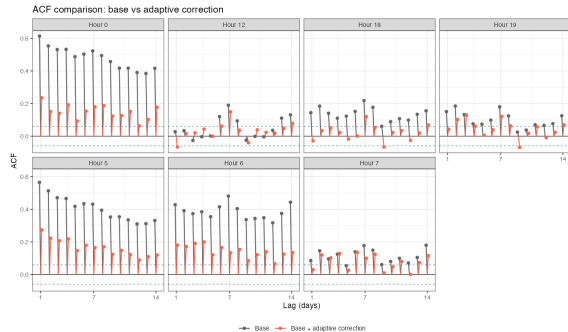
Where it loses, and the residual step



Positive = model beats official (MAE).

Hours 6–9: ramp timing is temperature-driven; at hr 8, MAPE loss $\approx 24\%$.

Errors are autocorrelated: Ljung–Box up to 7,227 at h6.



Correction:

$$\hat{Y}_d^{\text{res}} = \hat{Y}_d^{\text{base}} + \lambda_d \hat{r}_d$$

$\lambda_d, \hat{r}_d$ from strictly pre- d data (rolling CV) $\Rightarrow$ no leakage.

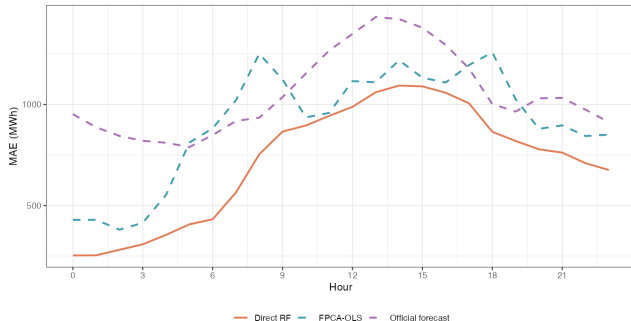
Result: +46.2% MAE, DM stat -71.2 .

Wins at all 24/24 hours.

Residual correction approach: (Xie and Hong, 2016)

Linearity: is the linear model adequate? (Sub-Q2)

MAE by hour: Direct RF vs FPCA-OLS (2023)



Main model: **OLS on FPCA scores**

Probe: direct RF predicting hourly load directly, bypassing FPCA entirely. Same features + lagged hourly values.

Model	MAE	vs. off.
Official (2025)	1052	ref.
FPCA-OLS (2025)	910	+13.5% ***
Direct RF (2025)	718	+31.8% ***

Reading (open): linear is adequate *within* the FPCA-score structure; the single-year RF gain hints at structure the FPCA compression discards.

Discussion and Conclusion

- **Expand the nonlinear probe to the full horizon.**

Direct RF's single-year gain (+31.8% on 2025 alone) is the clearest empirical lead. The natural next step is to extend the evaluation to the full 2023–2025 rolling-origin window and test whether the structural gain holds out of sample.

- **Develop the economic story of the underprediction.**

The ≈ 467 MWh/hour systematic bias is observed but unexplained. The thesis should characterise which curve segments drive it and connect this to the institutional context of the Turkish electricity market.

- **Complete the systematic literature review.**

FDA applied to electricity demand is an active area. Positioning the contribution requires a structured survey.

The core empirical results — FPCA representation, VAR pipeline, DM-confirmed accuracy, residual correction — are complete and validated. No new data collection required.

References

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